

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: DATA VISUALIZATION WITH POWER BI & TABLEAU

Practical No. 2

Aim: Working with Data Sources and Data Preparation in Tableau

- A. Connect to different data sources (Excel, CSV, Database)
- B. Create and use data extracts vs live connections
- C. Manage metadata (rename, hide, change data types)
- D. Perform joins (inner, left, right) on multiple tables
- E. Apply data blending between two sources
- F. Implement different types of filters (dimension, measure, date)
- G. Optimize performance using extracts and filtering

Dataset: E-commerce Dataset

Kaggle link:

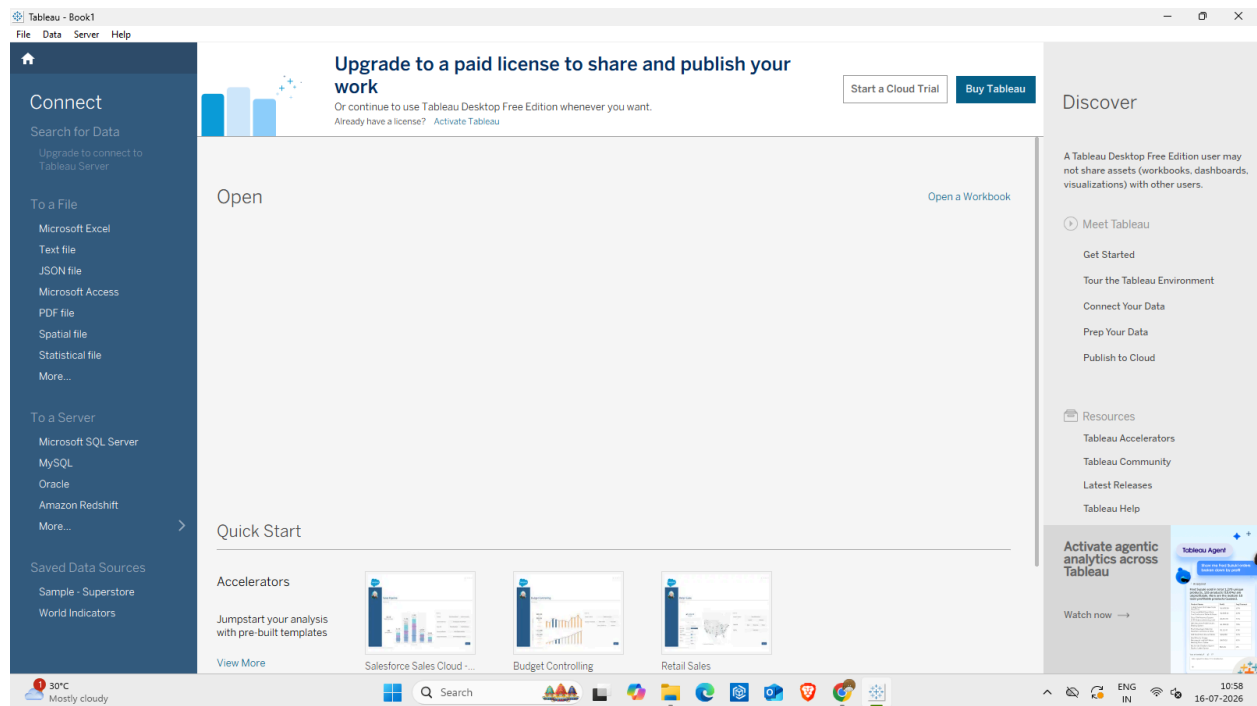
<https://www.kaggle.com/datasets/abdelrahmanalimo/e-commerce-dataset>

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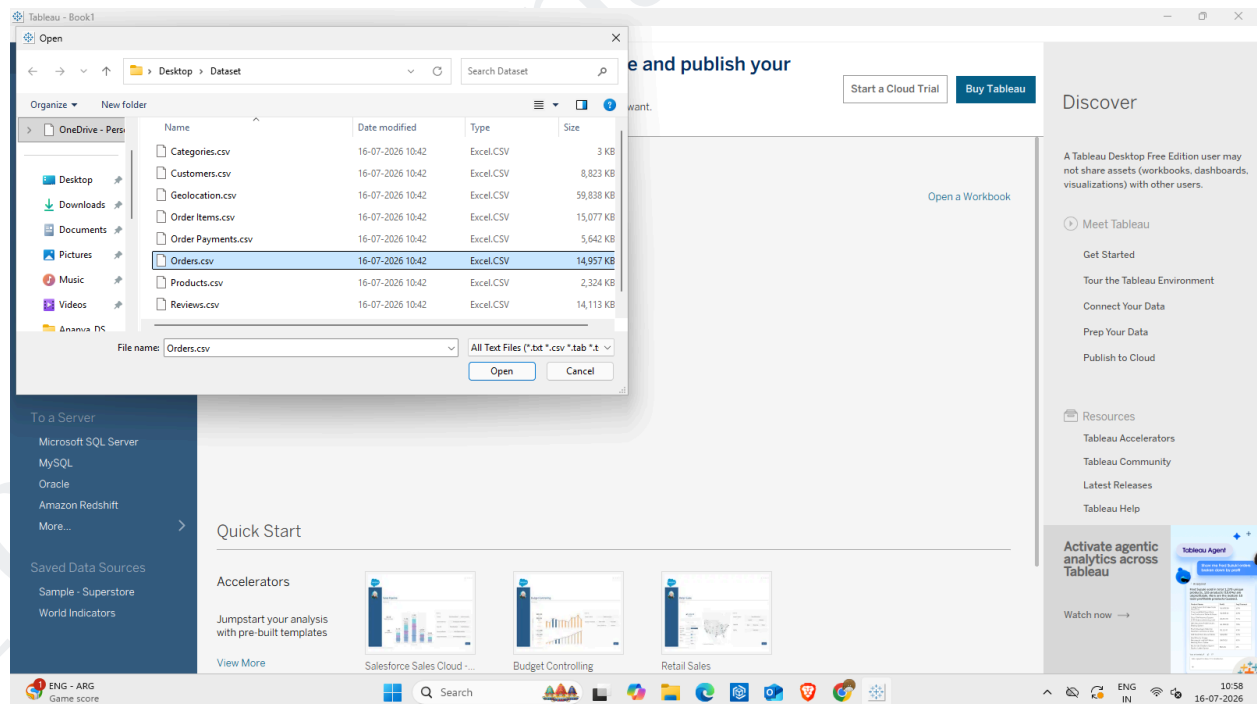
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Step A: Connect to different data sources (Excel, CSV, Database)

1. Launch Tableau Desktop on your machine



2. Go to **Connect** → **Text File**, select **Orders.csv**, and click **Open**.



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Step B: Creating Data Extracts vs. Live Connections

1. In the Data Source view, locate the connection mode toggle at the top-right corner and Switch the active selection from **Live** over to **Extract**.

Tableau - Book1

Connections: Orders (Text file)

Files: Use Data Interpreter, Categories.csv, Customers.csv, Geolocation.csv, Order Items.csv, Order Payments.csv, Orders.csv, Products.csv, Reviews.csv, Sellers.csv

Orders.csv

8 fields 99441 rows

Connection: ☒ Live ☐ Extract (selected) Edit Refresh Filters 0 Add

Extract will contain all data.

Order Id	Customer Id	Order Status	Order Purchase Timestamp	Order Approved At
91116032517d91c1180ea2bd...	cbece3180303b43f336eaab...	delivered	16-07-2017 09:50:00	16-07-2017 10:03:00
f5299da77d8dc1c9630692...	7ec3f2352e9adbb54e27488...	delivered	25-11-2017 13:50:00	28-11-2017 15:34:00
28c89c82267d381a04c5221...	32f6ea59402a367c36b5e5ec...	delivered	20-04-2017 21:43:00	25-04-2017 08:02:00
460bbd1de438f2781234032...	d47c07942091434330f8bf4...	delivered	22-06-2018 11:57:00	23-06-2018 01:40:00
24940d1052df86d8e24f16e...	dde3f6a475d39044e793c94...	delivered	01-05-2017 13:31:00	01-05-2017 13:50:00
9863811d92f6ae6fcd407813...	894344399c9fcb69219284f...	delivered	16-05-2018 13:53:00	16-05-2018 15:18:00
779078ac80552fa400b11c2...	f583831c572cc8569de2841f...	unavailable	13-06-2017 21:11:00	13-06-2017 21:22:00

2. Click **Sheet 1**, name the extract file **Ecommerce_Extract**, and click **Save**.

Tableau - Book1

Save Extract As

My Tableau Repository > Datasources

Organize: New folder

OneDrive - Pers

Name	Date modified	Type	Size
2026.2	23-06-2026 08:28	File folder	
Archive	23-06-2026 08:28	File folder	
Ecommerce_Extract	16-07-2026 11:00	Tableau Extract	11,008 KB

File name: Ecommerce_Extract

Save as type: Tableau Data Extract (*.hyper)

Save Cancel

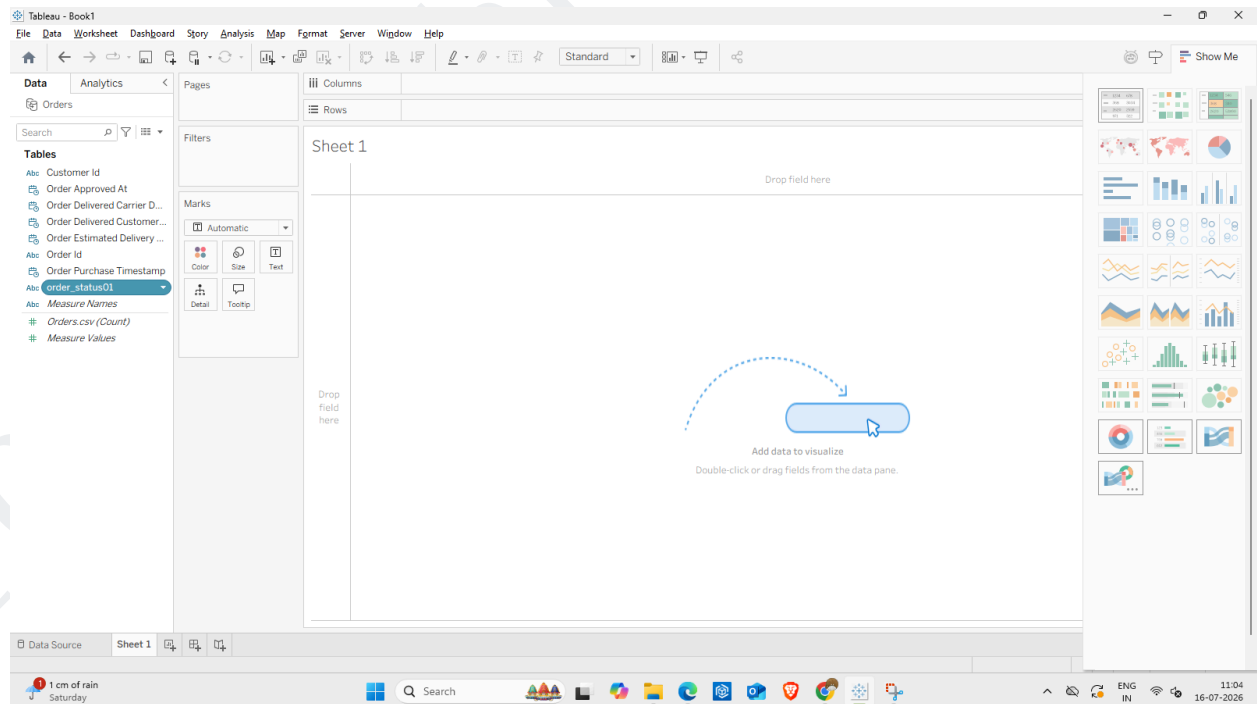
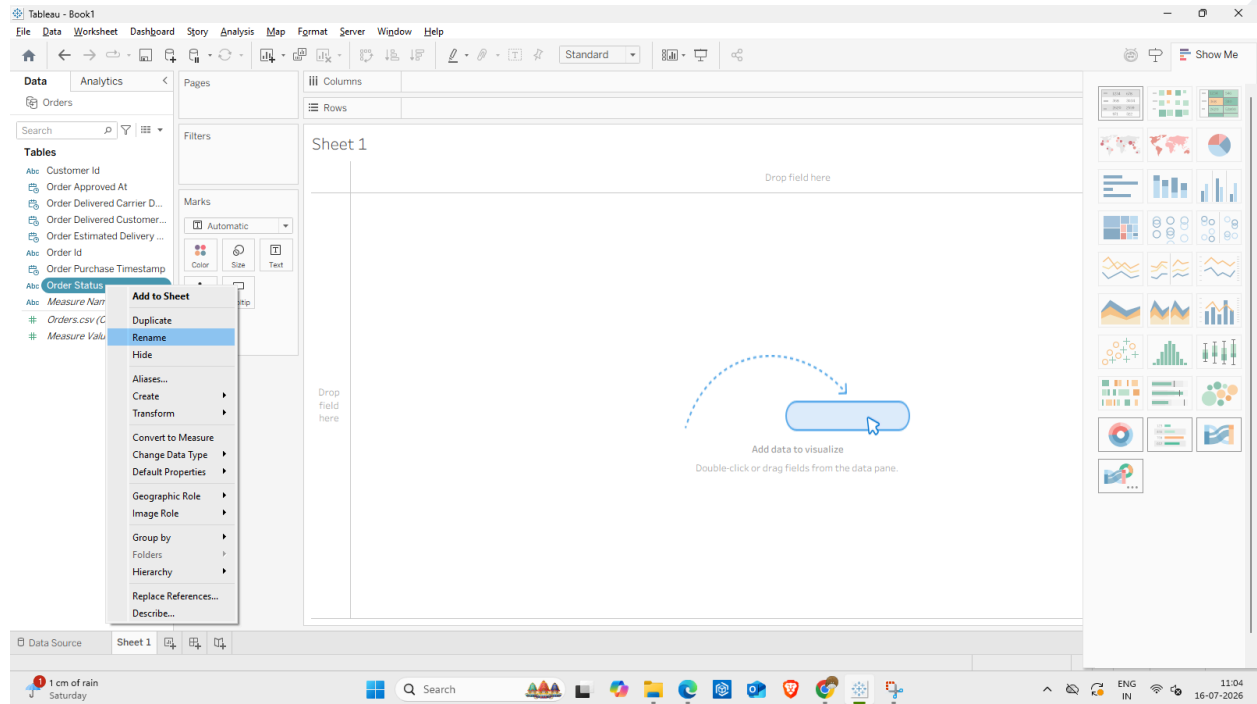
Tableau Desktop background showing the same data source view as the previous screenshot.

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Step C: Managing Operational Metadata

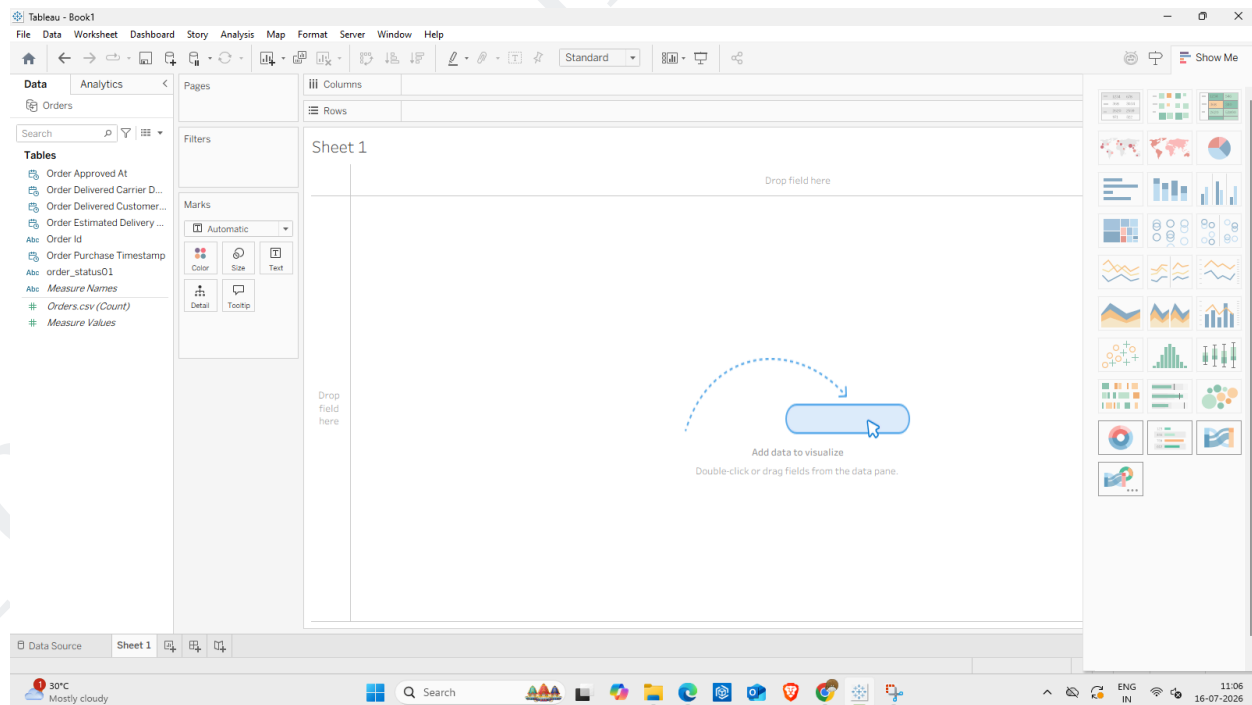
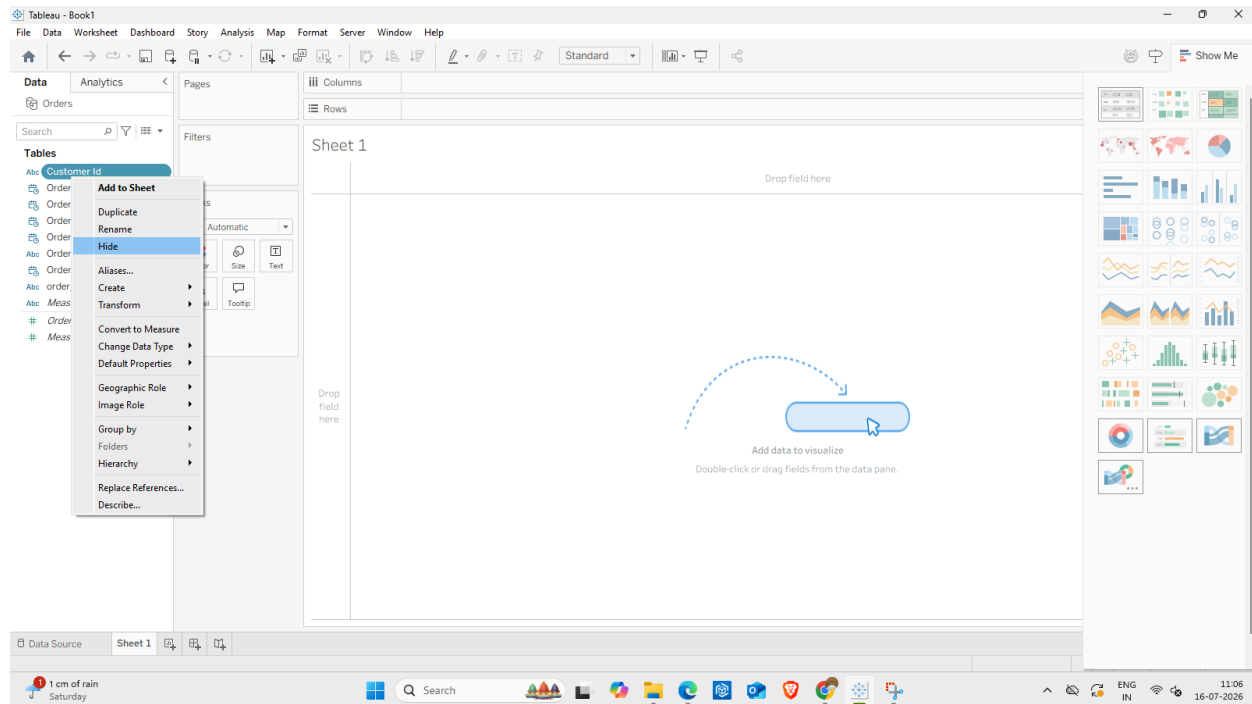
1. In **Sheet 1**, right-click **orders_status01** in the **Data Pane**, rename it to **Status**, and press **Enter**.



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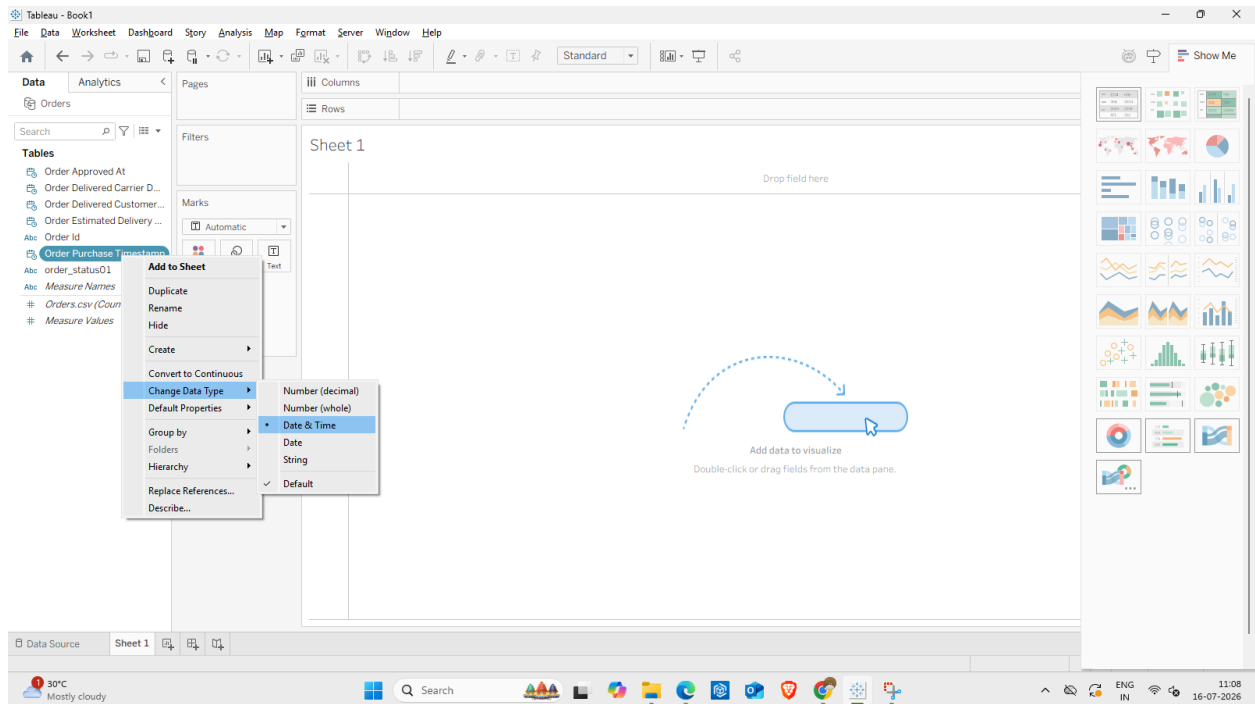
3. Hide a Field: Locate an internal tracking string variable that will not be used in visual reporting, such as **customer**. Right-click the field name and select **Hide**.



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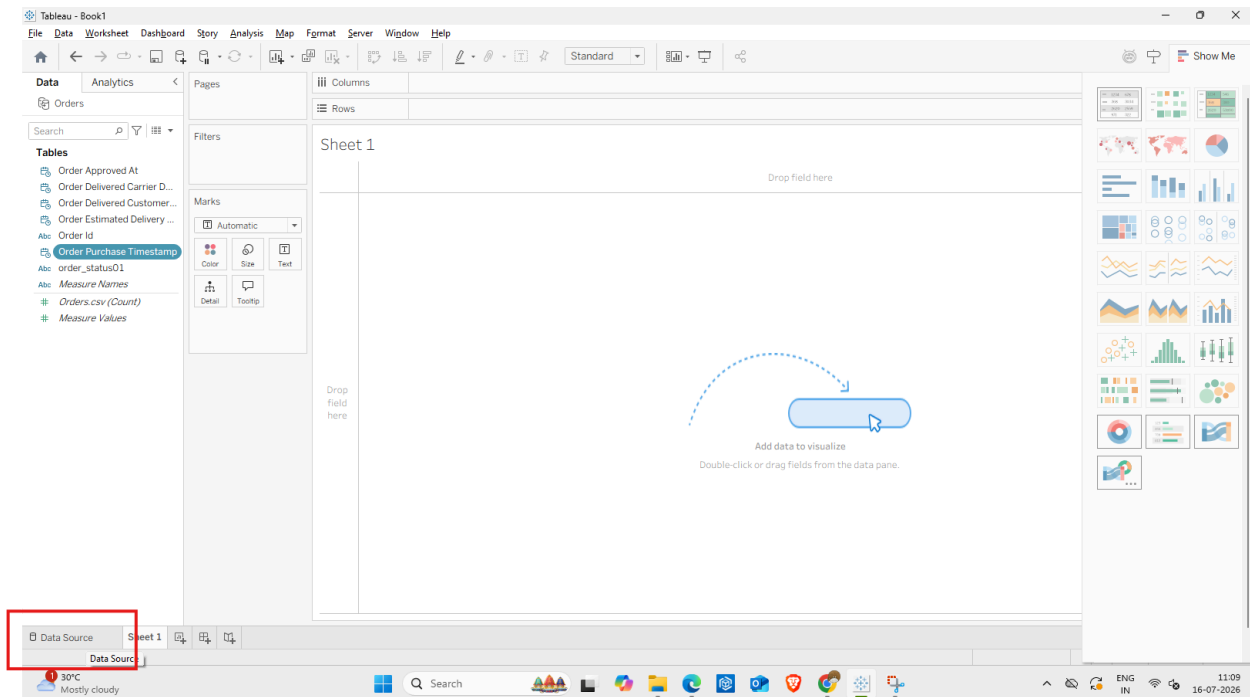
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4. Locate **order_purchase_timestamp**, click the data type icon beside it, and select **Date & Time** (if not already selected).

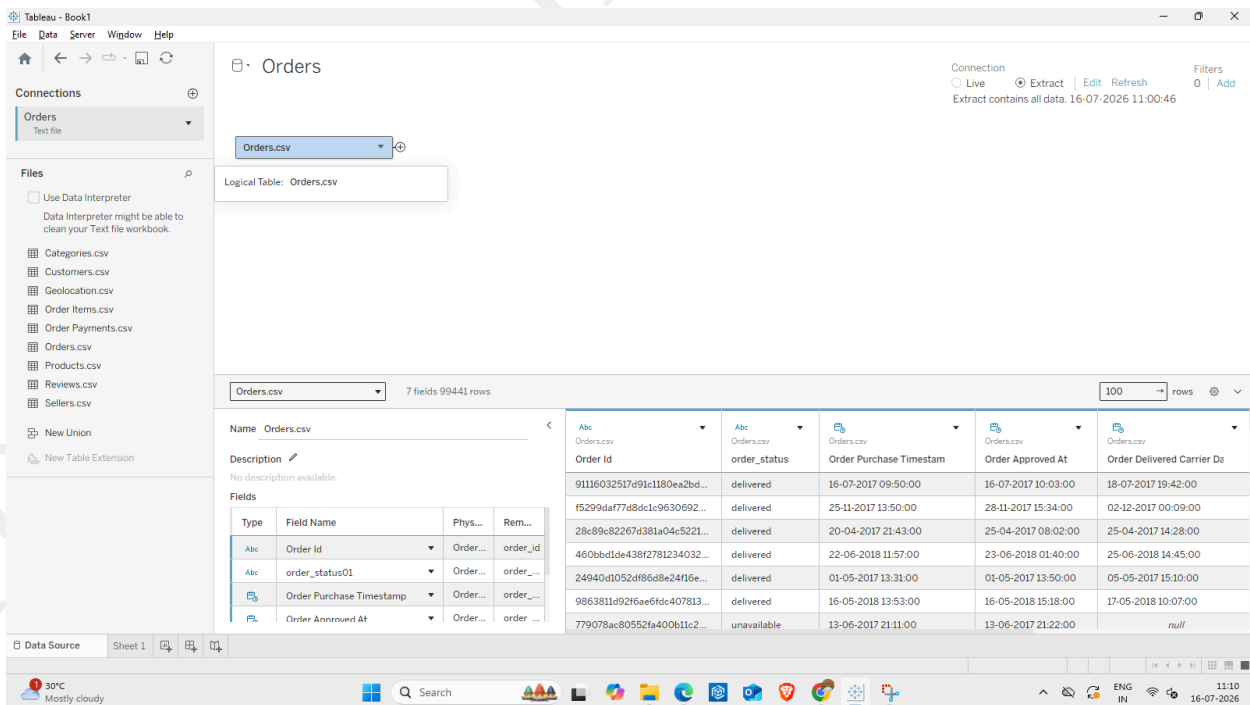


Step D: Performing Relational Joins on Multiple Tables

1. Click the Data Source tab positioned at the bottom-left corner of the window to return to the data schema designer.



2. In the top staging canvas area, locate the rectangular Orders.csv database block.



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3. Double-click the **Orders.csv** table to open the **Physical Layer**.

The screenshot shows the Tableau Desktop interface. On the left, the 'Connections' pane lists 'Orders' as a 'Text file'. The 'Files' pane shows a list of CSV files, including 'Orders.csv'. The main view displays the 'Orders' table in the 'Physical Layer'. A message box states 'Orders.csv is made of 1 table.' Below this, a table preview is shown with 7 fields and 99441 rows. The table has columns: Order Id, order_status, Order Purchase Timestamp, Order Approved At, and Order Delivered Carrier Da. The 'Fields' pane on the left shows the table's structure with columns: Type, Field Name, Phys..., and Rem....

Order Id	order_status	Order Purchase Timestamp	Order Approved At	Order Delivered Carrier Da
91116032517d91c180ea2bd...	delivered	16-07-2017 09:50:00	16-07-2017 10:03:00	18-07-2017 19:42:00
f5299da77d8dc1c9630692...	delivered	25-11-2017 13:50:00	28-11-2017 15:34:00	02-12-2017 00:09:00
28c89c82267d381a04c5221...	delivered	20-04-2017 21:43:00	25-04-2017 08:02:00	25-04-2017 14:28:00
460bbd1de438f2781234032...	delivered	22-06-2018 11:57:00	23-06-2018 01:40:00	25-06-2018 14:45:00
24940d1052df86d8e24f16e...	delivered	01-05-2017 13:31:00	01-05-2017 13:50:00	05-05-2017 15:10:00
9863811d92f6ae6f8c407813...	delivered	16-05-2018 13:53:00	16-05-2018 15:18:00	17-05-2018 10:07:00
779078ac80552fa400b11c2...	unavailable	13-06-2017 21:11:00	13-06-2017 21:22:00	null

4. Drag **Order Payments.csv** next to **Orders.csv** to create a join.

The screenshot shows the Tableau Desktop interface. On the left, the 'Connections' pane lists 'Orders' as a 'Text file'. The 'Files' pane shows a list of CSV files, including 'Order Payments.csv'. The main view displays the 'Orders' table in the 'Physical Layer'. A message box states 'Orders.csv is made of 2 tables.' Below this, a diagram shows a join between 'Orders.csv' and 'Order Payments.csv'. The table preview below shows 12 fields and 103886 rows. The table has columns: Order Id, order_status, Order Purchase Timestamp, Order Approved At, and Order Delivered Carrier Da. The 'Fields' pane on the left shows the table's structure with columns: Type, Field Name, Phys..., and Rem....

Order Id	order_status	Order Purchase Timestamp	Order Approved At	Order Delivered Carrier Da
b81ef226f3fe1789b1e8b2aca...	delivered	25-04-2018 22:01:00	25-04-2018 22:15:00	02-05-2018 15:20:00
a9810da82917af2d9aef1d27...	delivered	26-06-2018 11:01:00	26-06-2018 11:18:00	28-06-2018 14:18:00
25e8ea4e93996b6fa0d3dd7...	delivered	12-12-2017 11:19:00	14-12-2017 09:52:00	15-12-2017 20:13:00
ba78997921bbcd1373bb41e...	delivered	06-12-2017 12:04:00	06-12-2017 12:13:00	07-12-2017 20:28:00
42f8f880ba16b47b5925dd4...	delivered	21-05-2018 13:59:00	21-05-2018 16:14:00	22-05-2018 11:46:00
298fcd1f73eb413e4d26d01...	delivered	07-05-2018 13:20:00	07-05-2018 15:31:00	10-05-2018 13:35:00
771ee386b001f06208a7419e...	delivered	23-06-2017 13:11:00	23-06-2017 13:25:00	26-06-2017 17:33:00

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5. Click directly on the **Venn diagram icon** to open the Join configuration properties box.

Tableau - Book1

File Data Server Window Help

Connections

Orders
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Categories.csv
Customers.csv
Geolocation.csv
Order Items.csv
Order Payments.csv
Orders.csv
Products.csv
Reviews.csv
Sellers.csv

New Union
New Table Extension

Orders

Orders.csv is made of 2 tables. ⓪

Orders.csv

Order Payments.csv

Join

Inner Left Right Full Outer

Data Source

Order Id = order id (Order ...)

Add new join clause

Orders.csv 12 fields 103886 rows

100 rows

Name Orders.csv

Description

No description available.

Fields

Type	Field Name	Phys...	Rem...
Abc	Order Id	Order...	order_id
Abc	order_status01	Order...	order_...
Abc	Order Purchase Timestamp	Order...	order_...
Abc	Order Approved At	Order...	order_...

Abc Orders.csv Order Id	Abc Orders.csv order_status	Abc Orders.csv Order Purchase Timestamp	Abc Orders.csv Order Approved At	Abc Orders.csv Order Delivered Carrier Da
b81ef226f3fe1789be8b2aca...	delivered	25-04-2018 22:01:00	25-04-2018 22:15:00	02-05-2018 15:20:00
a9810da82917af2d9aefd127...	delivered	26-06-2018 11:01:00	26-06-2018 11:18:00	28-06-2018 14:18:00
25e8ea4e93396b6fa0d3dd7...	delivered	12-12-2017 11:19:00	14-12-2017 09:52:00	15-12-2017 20:13:00
ba78997921bbcd1373bb41e...	delivered	06-12-2017 12:04:00	06-12-2017 12:13:00	07-12-2017 20:28:00
42fdf880ba16b47b5925dd4...	delivered	21-05-2018 13:59:00	21-05-2018 16:14:00	22-05-2018 11:46:00
298fcd1f73eb413e4d26d01...	delivered	07-05-2018 13:20:00	07-05-2018 15:31:00	10-05-2018 13:35:00
771ee386b00106208a7419e...	delivered	23-06-2017 13:11:00	23-06-2017 13:25:00	26-06-2017 17:33:00

Data Source Sheet 1

NIFTY +0.34%

Search

ENG IN

11:12 16-07-2026

6. Set **order_id** as the matching key for both tables (if not already selected).

Tableau - Book1

File Data Server Window Help

Connections

Orders
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Categories.csv
Customers.csv
Geolocation.csv
Order Items.csv
Order Payments.csv
Orders.csv
Products.csv
Reviews.csv
Sellers.csv

New Union
New Table Extension

Orders

Orders.csv is made of 2 tables. ⓪

Orders.csv

Order Payments.csv

Join

Inner Left Right Full Outer

Data Source

Order Id = order id (Order ...)

Add new join clause

Orders.csv 12 fields 103886 rows

100 rows

Name Orders.csv

Description

No description available.

Fields

Type	Field Name	Phys...	Rem...
Abc	Order Id	Order...	order_id
Abc	order_status01	Order...	order_...
Abc	Order Purchase Timestamp	Order...	order_...
Abc	Order Approved At	Order...	order_...

Abc Orders.csv Order Id	Abc Orders.csv order_status	Abc Orders.csv Order Purchase Timestamp	Abc Orders.csv Order Approved At	Abc Orders.csv Order Delivered Carrier Da
b81ef226f3fe1789be8b2aca...	delivered	25-04-2018 22:01:00	25-04-2018 22:15:00	02-05-2018 15:20:00
a9810da82917af2d9aefd127...	delivered	26-06-2018 11:01:00	26-06-2018 11:18:00	28-06-2018 14:18:00
25e8ea4e93396b6fa0d3dd7...	delivered	12-12-2017 11:19:00	14-12-2017 09:52:00	15-12-2017 20:13:00
ba78997921bbcd1373bb41e...	delivered	06-12-2017 12:04:00	06-12-2017 12:13:00	07-12-2017 20:28:00
42fdf880ba16b47b5925dd4...	delivered	21-05-2018 13:59:00	21-05-2018 16:14:00	22-05-2018 11:46:00
298fcd1f73eb413e4d26d01...	delivered	07-05-2018 13:20:00	07-05-2018 15:31:00	10-05-2018 13:35:00
771ee386b00106208a7419e...	delivered	23-06-2017 13:11:00	23-06-2017 13:25:00	26-06-2017 17:33:00

Data Source Sheet 1

NIFTY +0.34%

Search

ENG IN

11:13 16-07-2026

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The screenshot shows the Tableau interface with a worksheet named 'Orders'. A join dialog box is open, showing 'Orders.csv' and 'Order Payments.csv' joined by an Inner Join. The join condition is 'Order Id' from 'Orders.csv' equals 'order id (Order ...)' from 'Order Payments.csv'. The data preview shows the following data:

Order Id	order_status	Order Purchase Timestamp	Order Approved At	Order Delivered Carrier Ds
b81ef226f3fe1789b1e8b2aca...	delivered	25-04-2018 22:01:00	25-04-2018 22:15:00	02-05-2018 15:20:00
a9810da82917af2d9aefd127...	delivered	26-06-2018 11:01:00	26-06-2018 11:18:00	28-06-2018 14:18:00
25e8ea4e93396b6fa0d3dd7...	delivered	12-12-2017 11:19:00	14-12-2017 09:52:00	15-12-2017 20:13:00
ba78997921bbcd1373bb41e...	delivered	06-12-2017 12:04:00	06-12-2017 12:13:00	07-12-2017 20:28:00
42fd880ba16b47b5925dd4...	delivered	21-05-2018 13:59:00	21-05-2018 16:14:00	22-05-2018 11:46:00
298fcd1f73eb413e4d26d01...	delivered	07-05-2018 13:20:00	07-05-2018 15:31:00	10-05-2018 13:35:00
771ee386b00106208a7419e...	delivered	23-06-2017 13:11:00	23-06-2017 13:25:00	26-06-2017 17:33:00

7. Select **Inner Join** and observe the data preview, which displays only the records with matching **order_id** values in both tables.

The screenshot shows the same Tableau interface as the previous one, but with the 'Add new join clause' button visible in the join dialog. The data preview is identical to the previous one.

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8. Leave the configuration set to a **Left Join** to preserve complete order history logs for the remainder of this exercise.

The screenshot shows the Tableau Desktop interface. On the left, the 'Connections' pane shows 'Orders' as a text file. The 'Files' pane lists various CSV files including 'Categories.csv', 'Customers.csv', 'Geolocation.csv', 'Order Items.csv', 'Order Payments.csv', 'Orders.csv', 'Products.csv', 'Reviews.csv', and 'Sellers.csv'. The main workspace displays a join configuration for 'Orders.csv' and 'Order Payments.csv'. The join type is set to 'Left', which is highlighted. A tooltip explains: 'Includes all values in left table and all matches from right table. Members without matches will show up as nulls on the right.' Below the join configuration, a data preview table is shown with columns: 'Order Id', 'order_status', 'Order Purchase Timestamp', 'Order Approved At', and 'Order Delivered Carrier Da'. The table contains 10 rows of data.

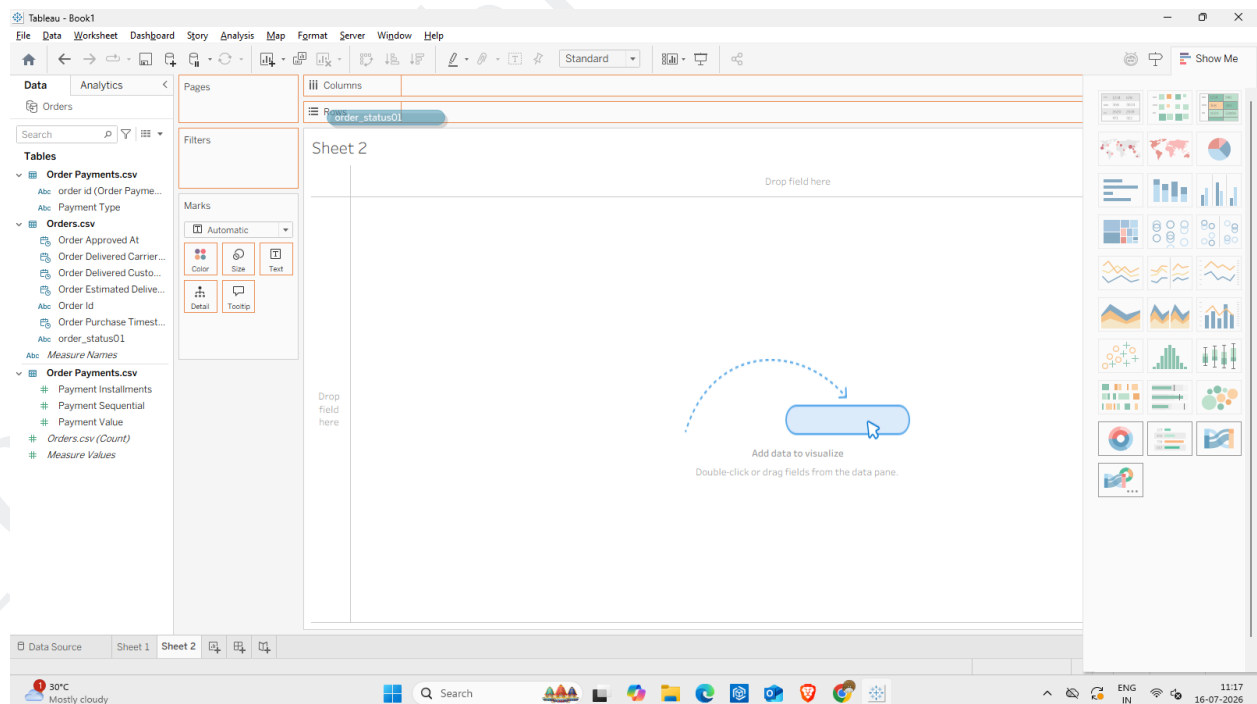
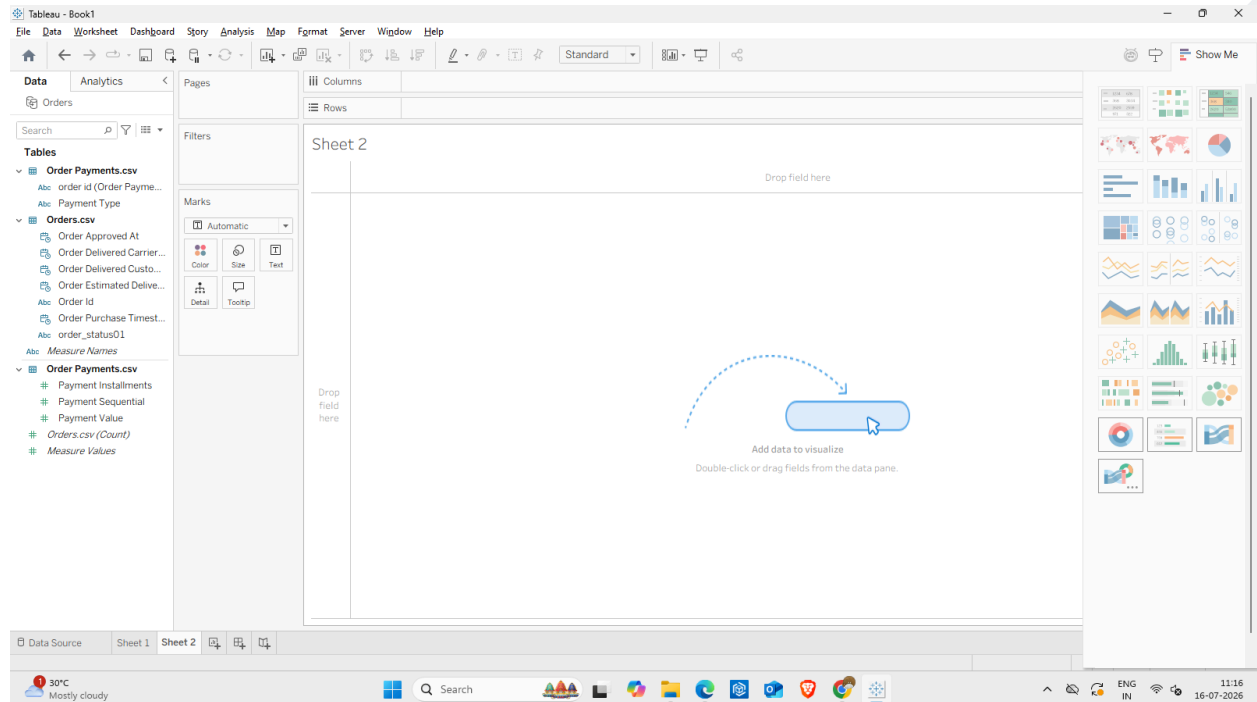
Order Id	order_status	Order Purchase Timestamp	Order Approved At	Order Delivered Carrier Da
b81ef226f3fe1789b1e8b2aca...	delivered	25-04-2018 22:01:00	25-04-2018 22:15:00	02-05-2018 15:20:00
a9810da82917af2d9aefd127...	delivered	26-06-2018 11:01:00	26-06-2018 11:18:00	28-06-2018 14:18:00
25e8ea4e93396b6fa0d3dd7...	delivered	12-12-2017 11:19:00	14-12-2017 09:52:00	15-12-2017 20:13:00
ba78997921bbcdc1373bb41e...	delivered	06-12-2017 12:04:00	06-12-2017 12:13:00	07-12-2017 20:28:00
42fdf880ba16b47b5925idd4...	delivered	21-05-2018 13:59:00	21-05-2018 16:14:00	22-05-2018 11:46:00
298fcd1f73eb413e4d26d01...	delivered	07-05-2018 13:20:00	07-05-2018 15:31:00	10-05-2018 13:35:00
771ee386b001f06208a7419e...	delivered	23-06-2017 13:11:00	23-06-2017 13:25:00	26-06-2017 17:33:00

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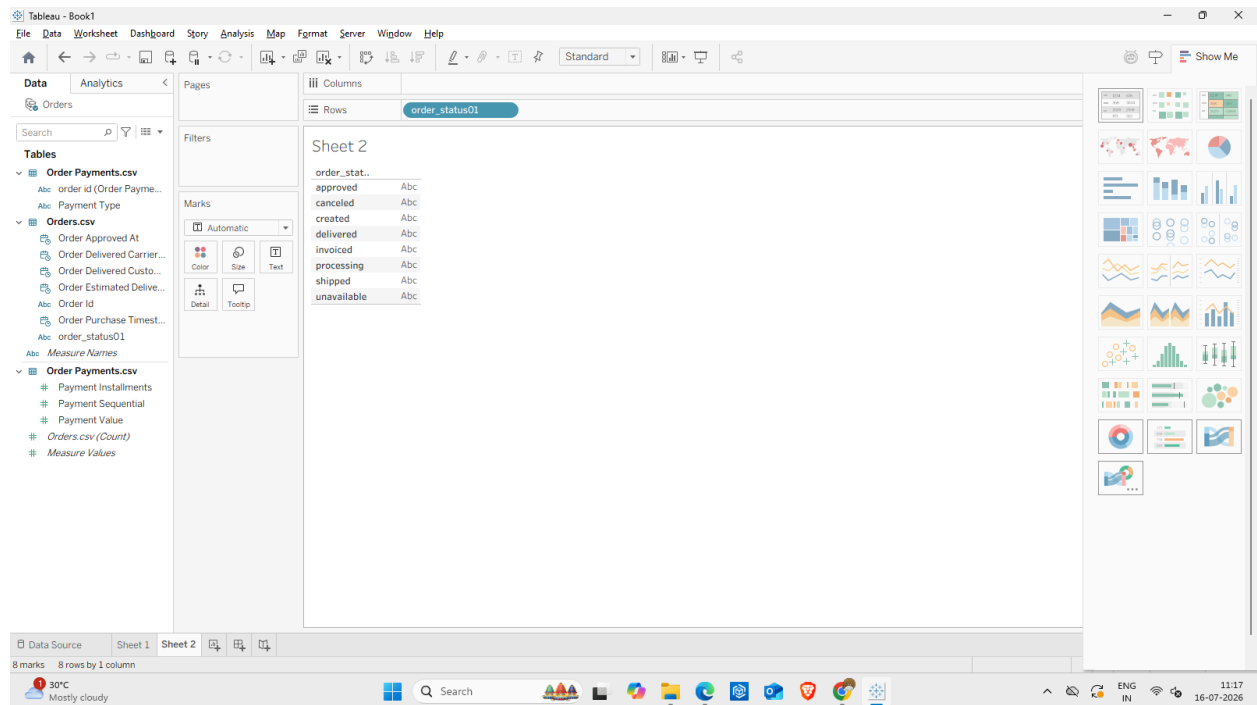
Step E: Applying Data Blending Between Independent Sources

1. Create a new worksheet by clicking the **New Worksheet (+)** icon to open **Sheet 2**. Select the joined **Orders** dataset, then drag the modified **Status** dimension to the **Rows** shelf. A blue checkmark indicates it is the **Primary** data pipeline.

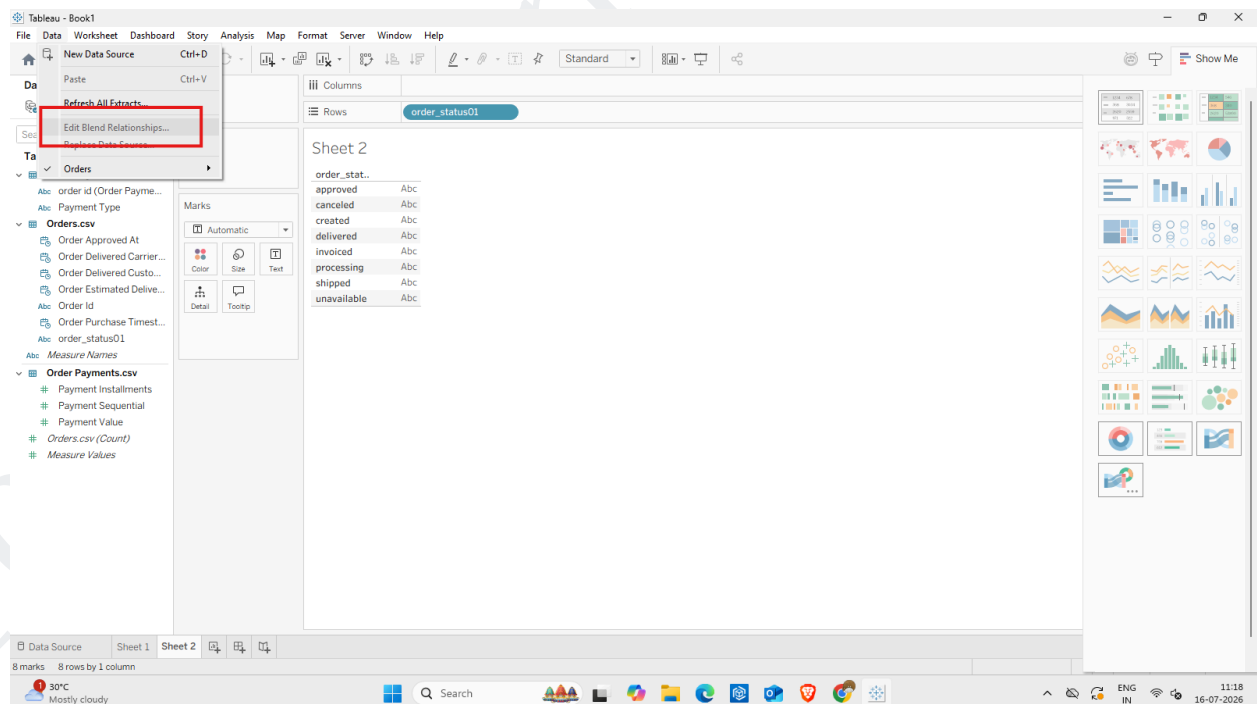


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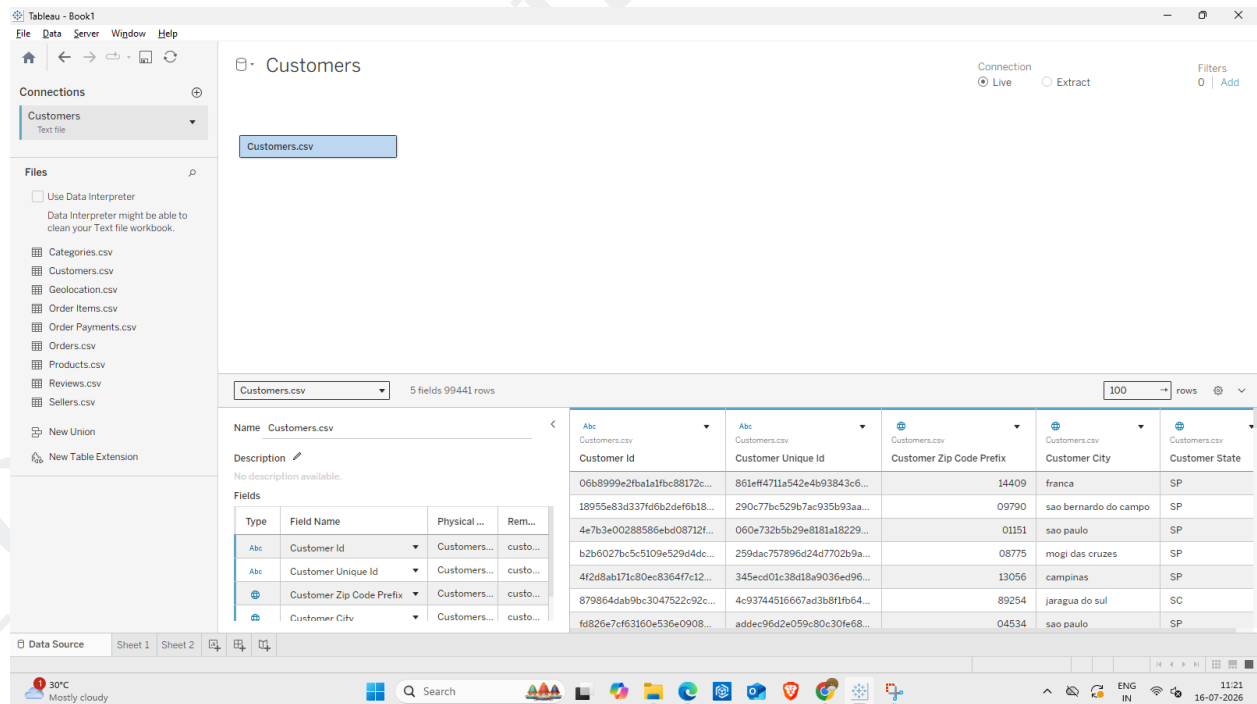
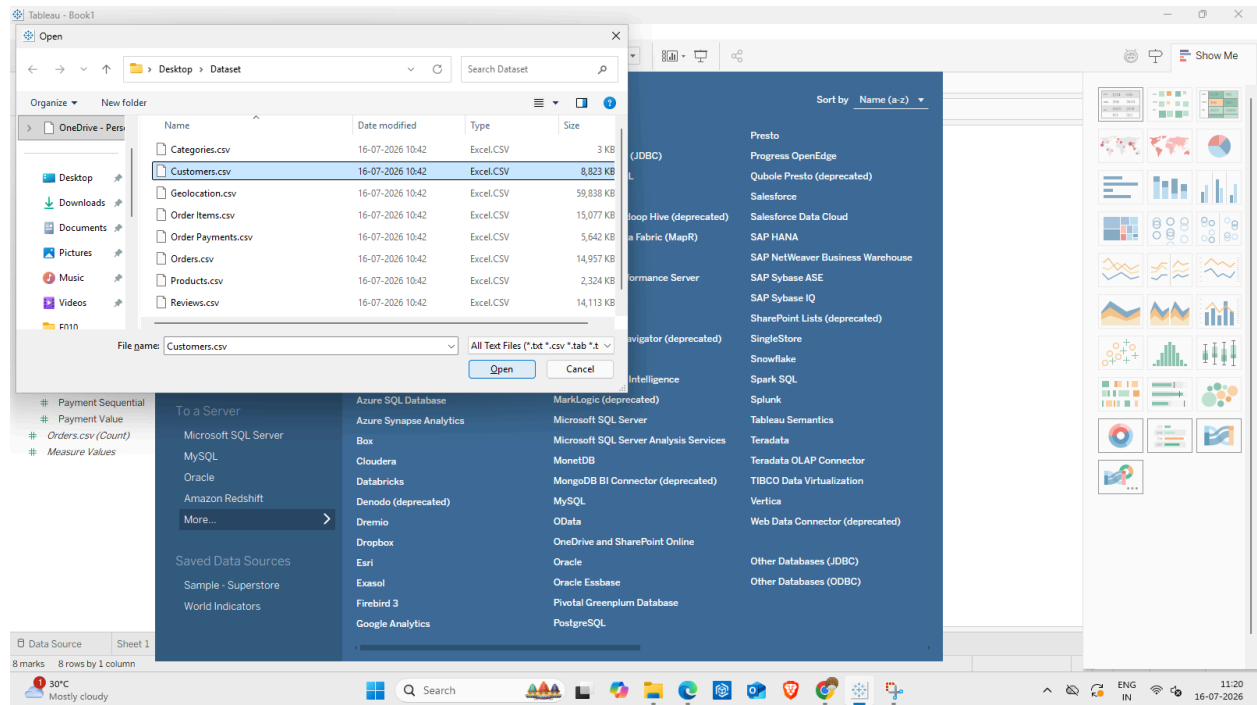
2. From the top menu bar, select **Data** → **Edit Blend Relationships**. If the option is grayed out, first click **New Data Source**, then proceed to **Edit Blend Relationships**.



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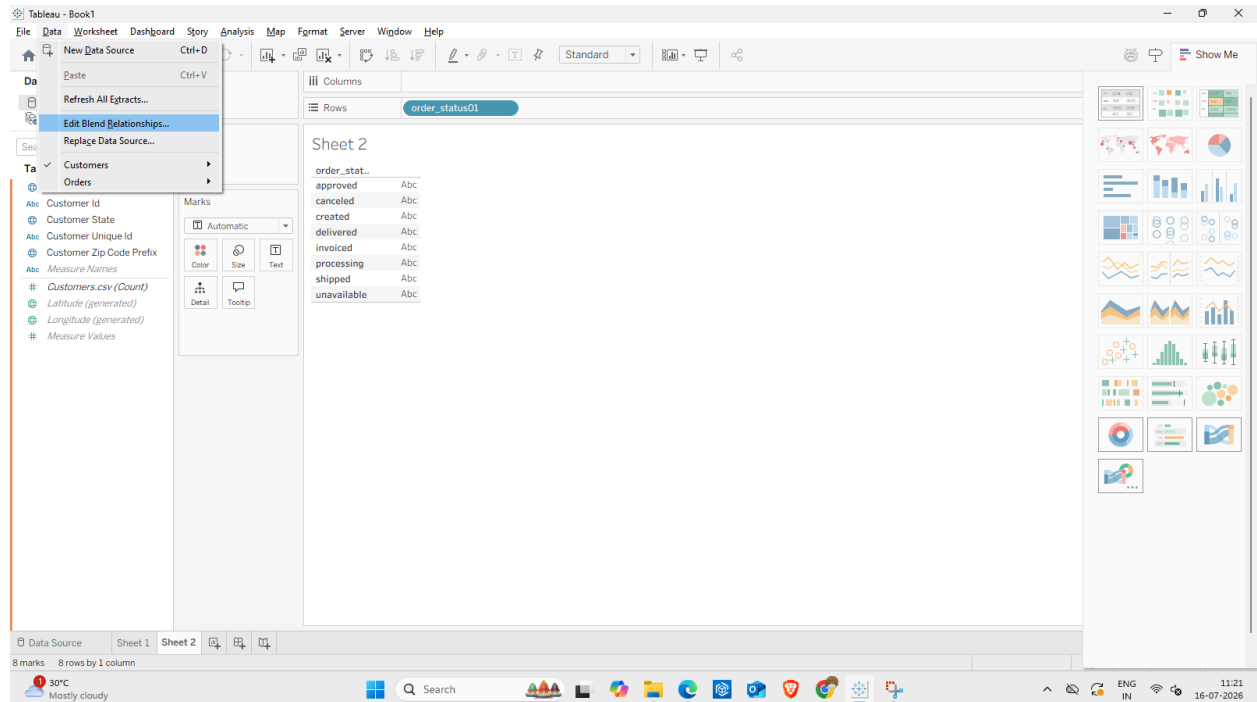
3. After clicking **New Data Source**, the **Data Source** window opens. Select **Text File** and import the **Customers.csv** file.



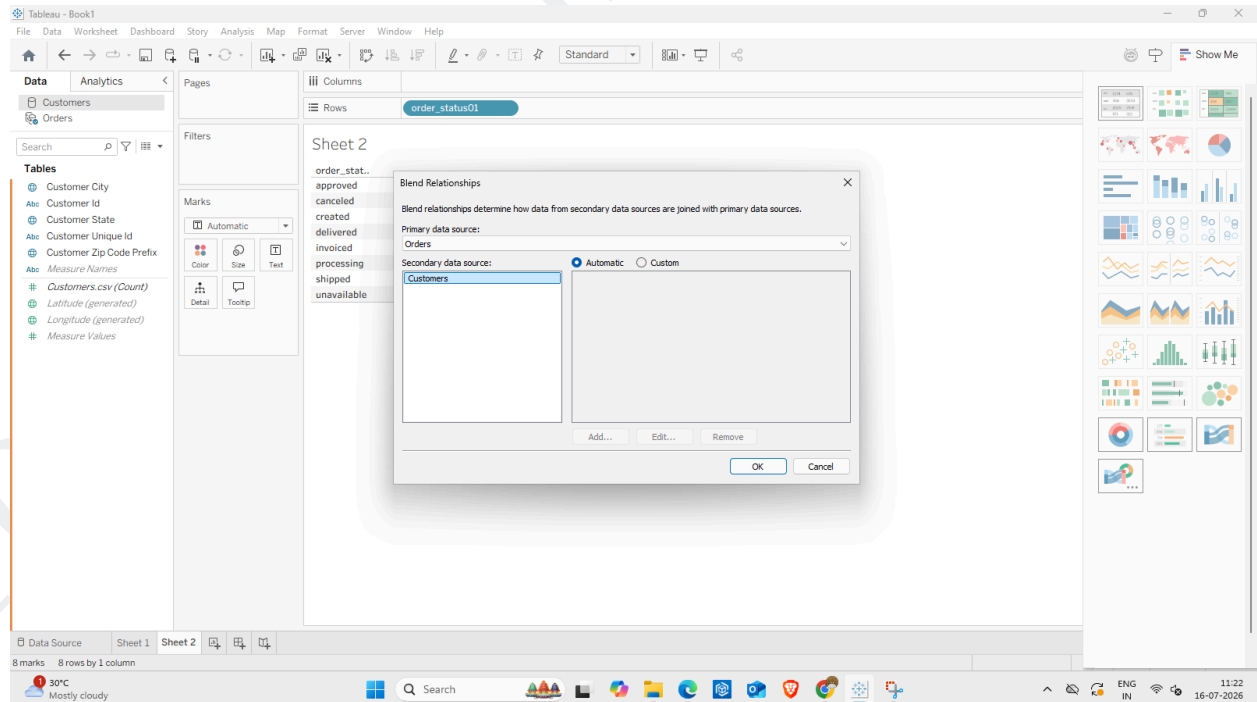
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4. Return to **Sheet 2** and click **Data** → **Edit Blend Relationships**.



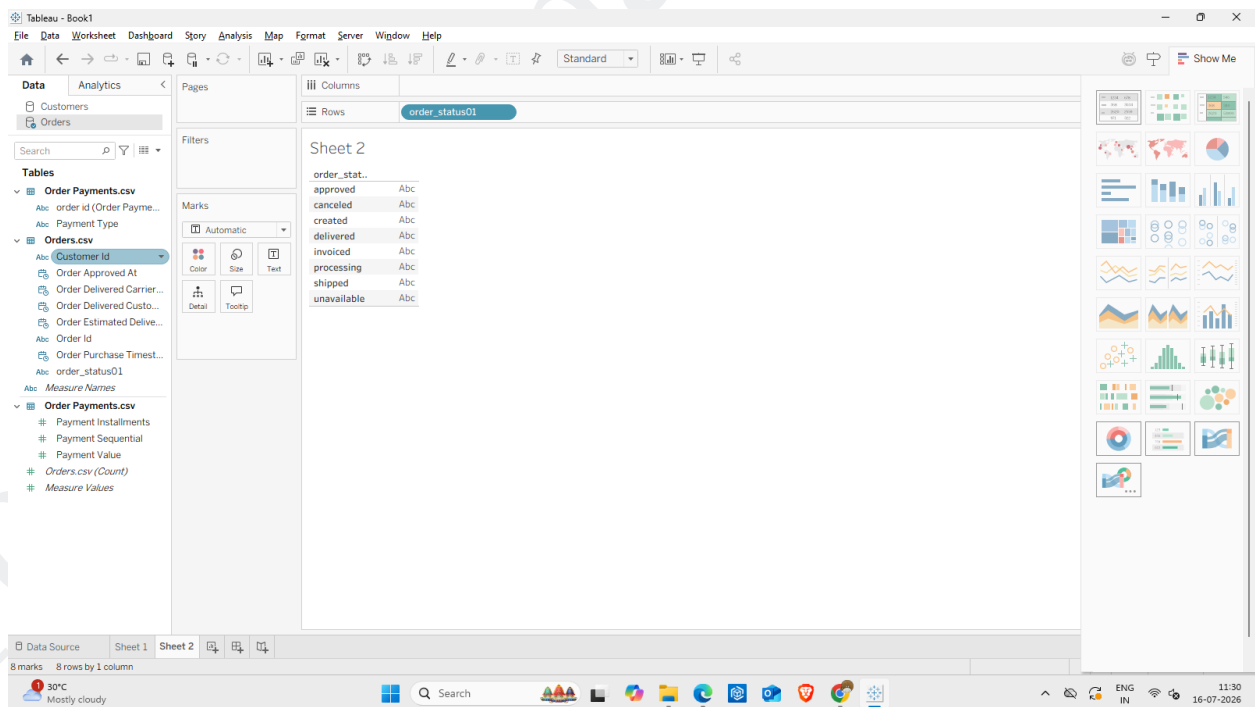
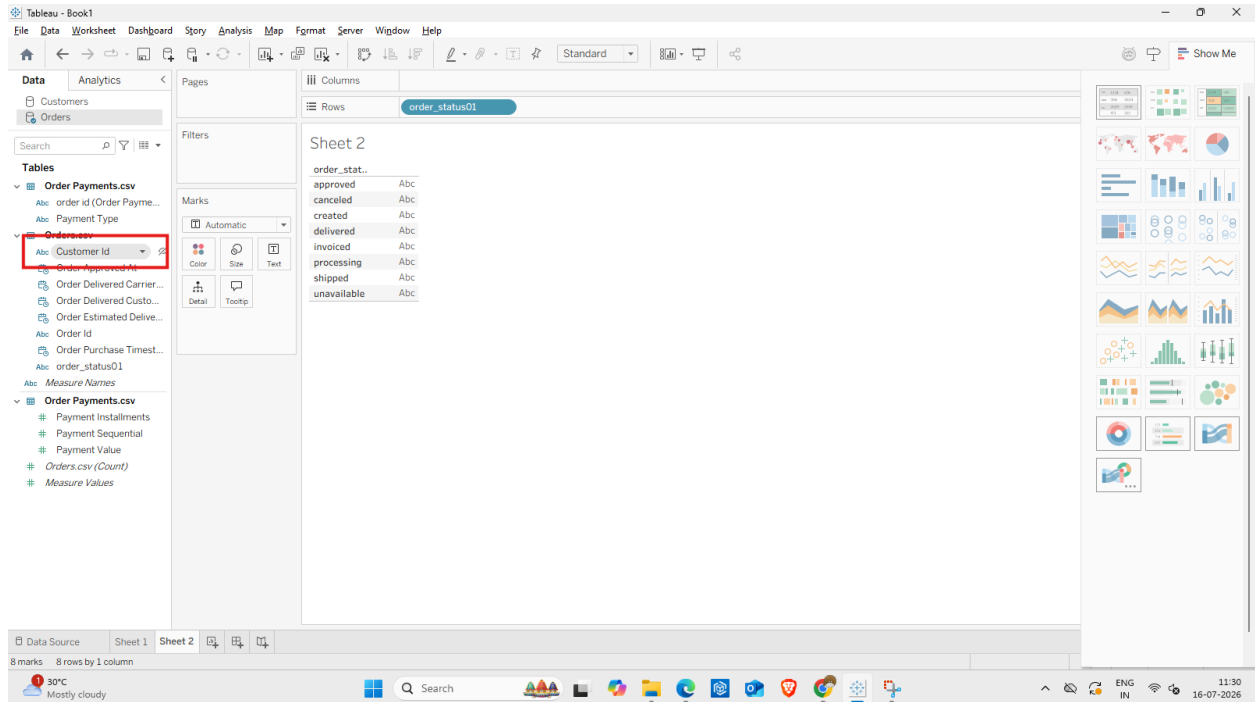
5. Set the blend relationship to **Custom**, click **Add**, match **customer_id** with **customer_id**, and click **OK**.



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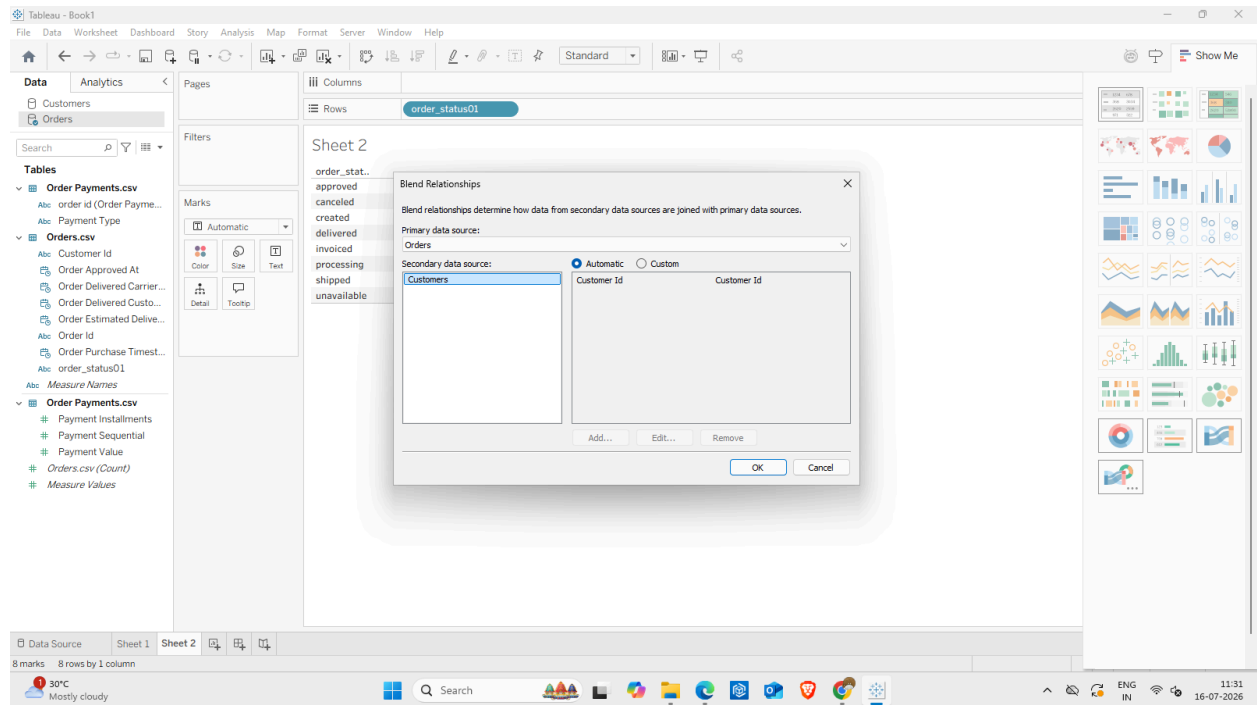
After Clicking it the **Customer Id** comes back into the data pane and make sure you click the “**eye button**” next to it marked in bellow provided snapshot



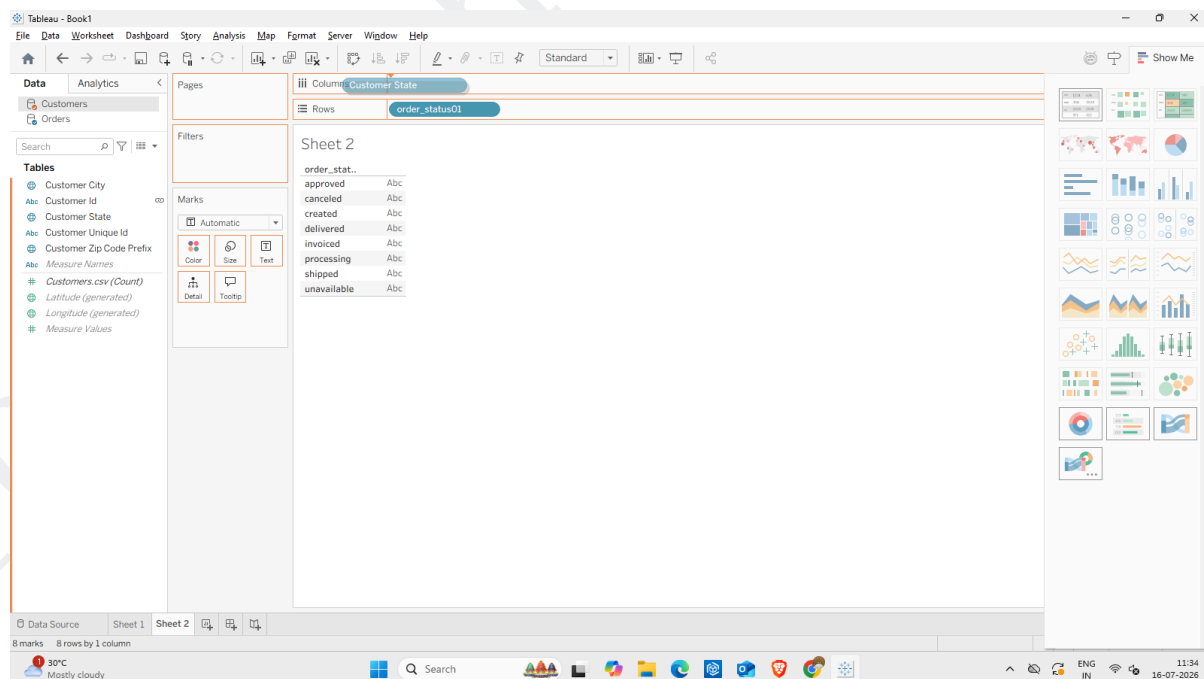
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Return to **Data** → **Edit Blend Relationships**, ensure **customer_id** appears in both datasets, select **Custom**, and click **OK**.



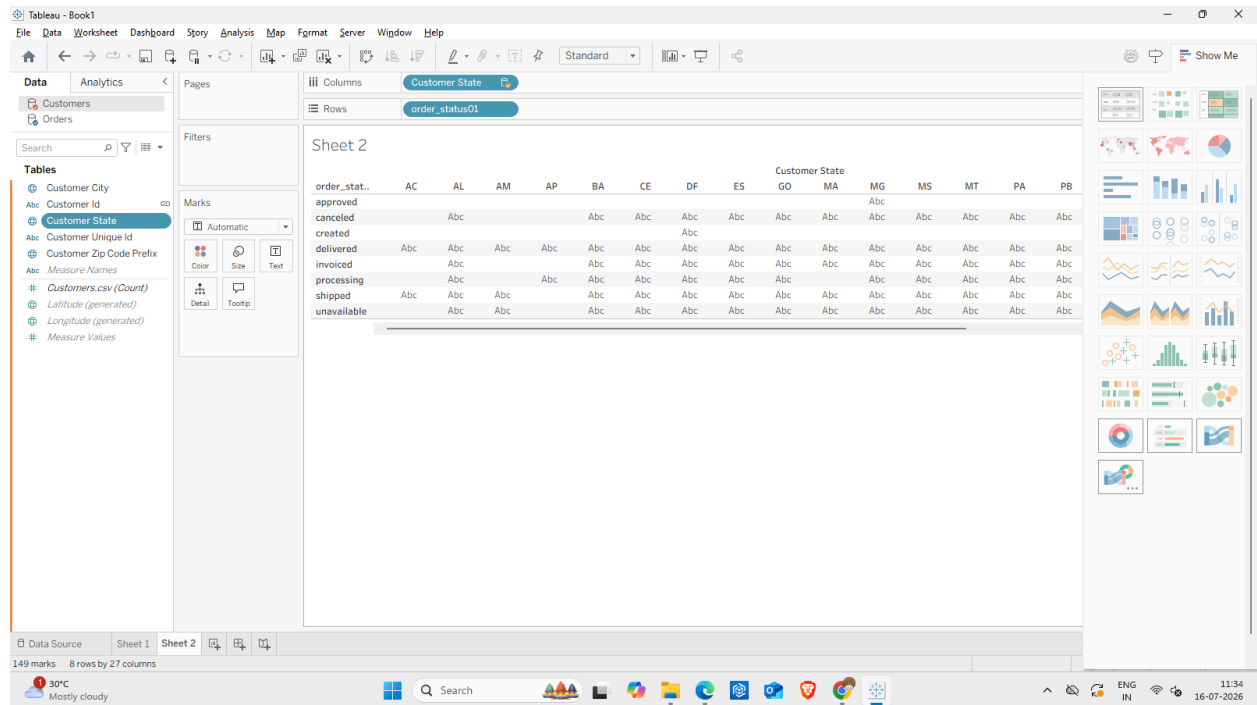
6. In the **Data Pane**, select the **Customers** dataset. An orange chain icon next to **customer_id** confirms the datasets are linked.



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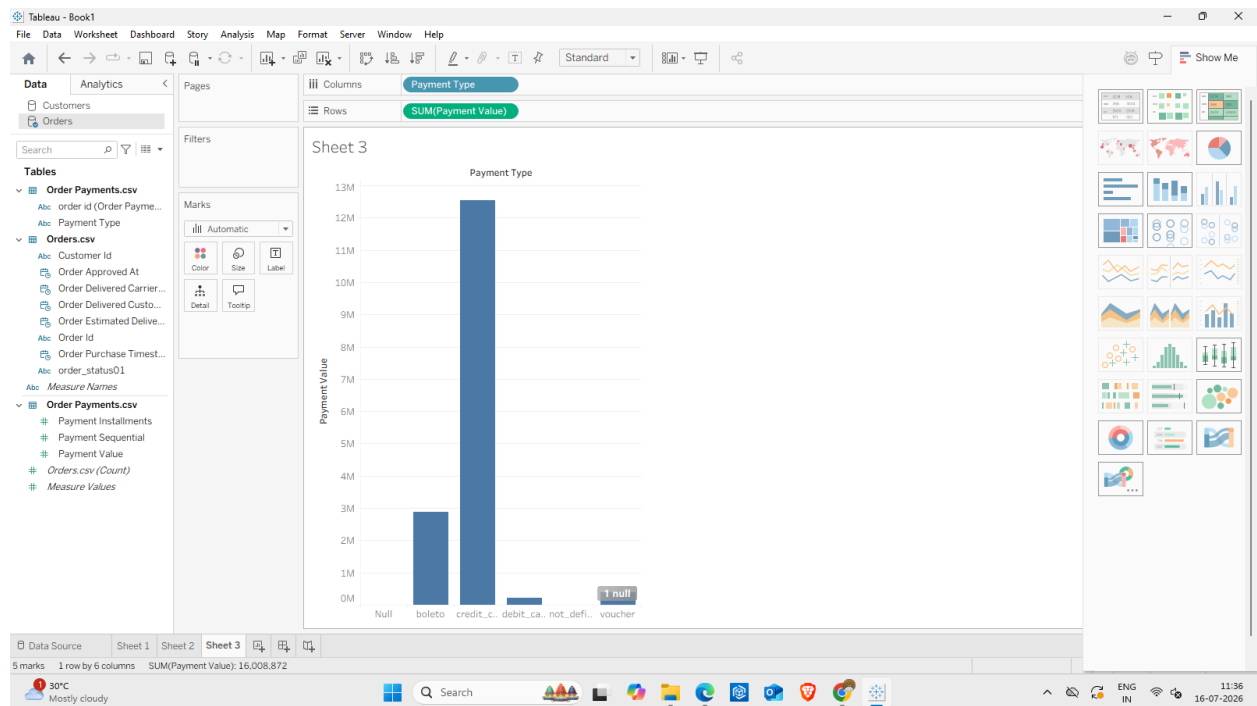
7. Drag the **customer_state** dimension from the **Customers** dataset to the **Columns** shelf.



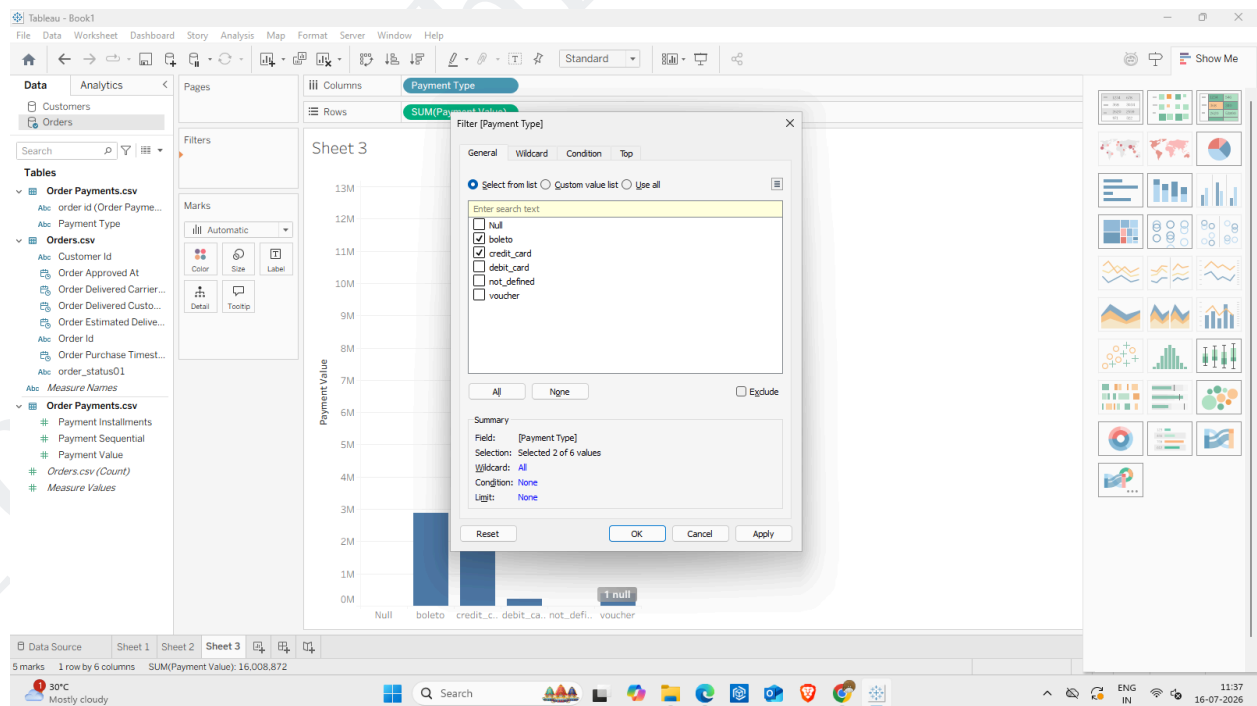
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Step F: Implementing Multi-Tiered Filters 1. Open a new worksheet, then drag **payment_value** to the **Rows** shelf and **payment_type** to the **Columns** shelf.



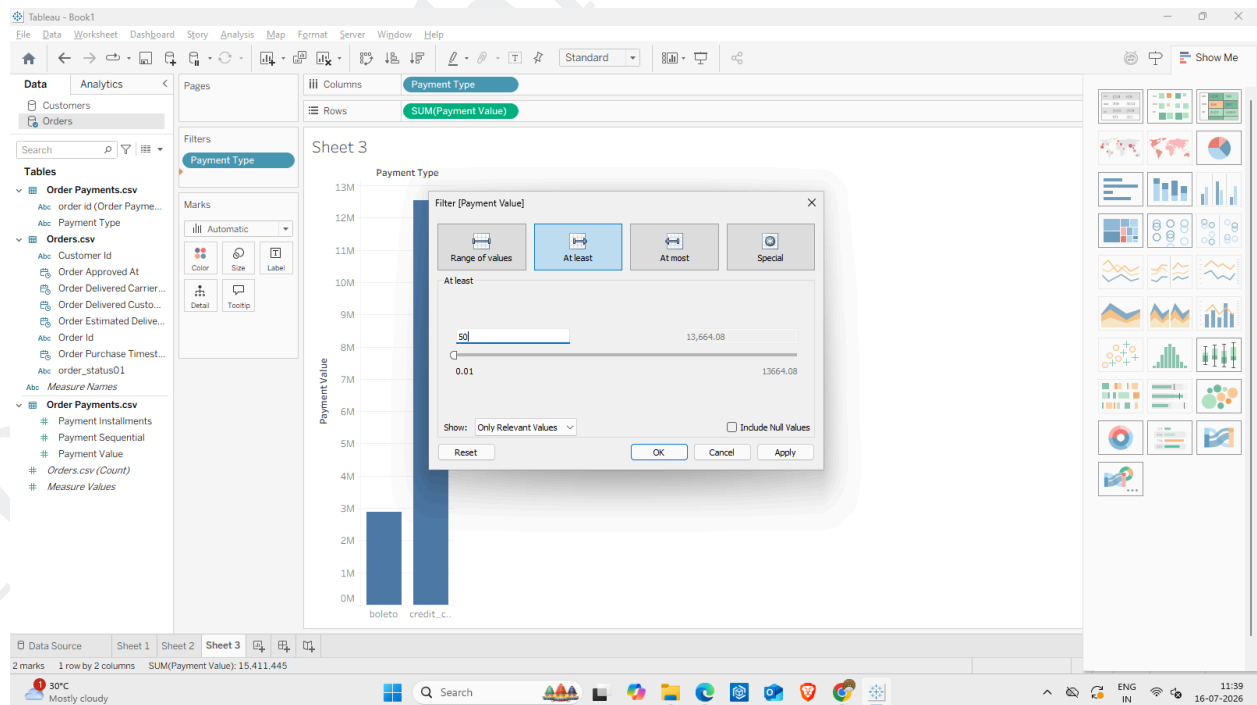
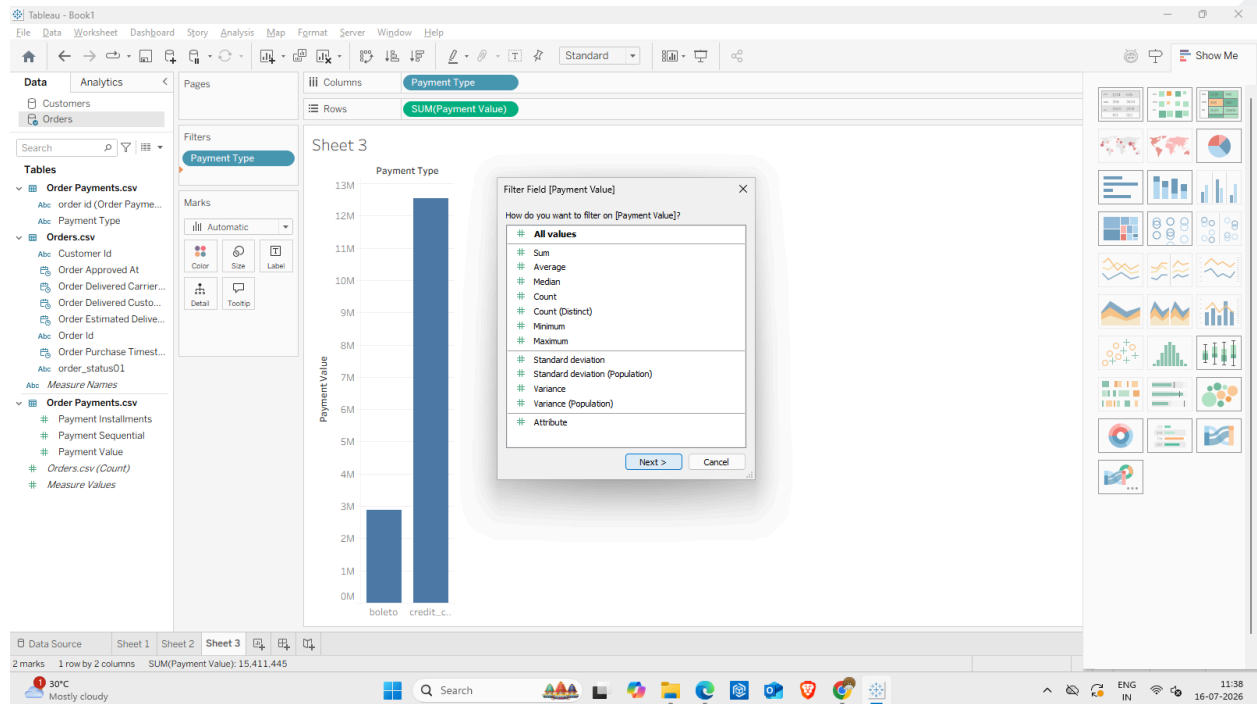
2. Drag **payment_type** to the **Filters** shelf, select only **credit_card** and **boleto**, deselect **not_defined**, and click **OK**.



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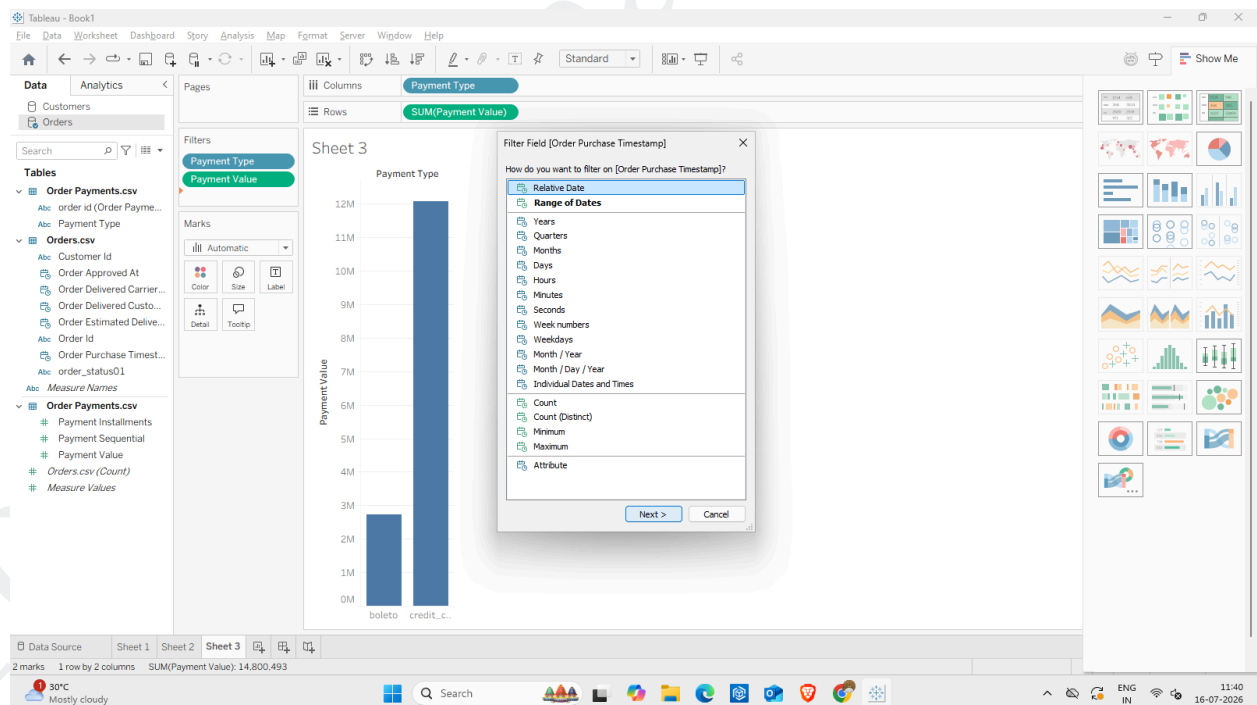
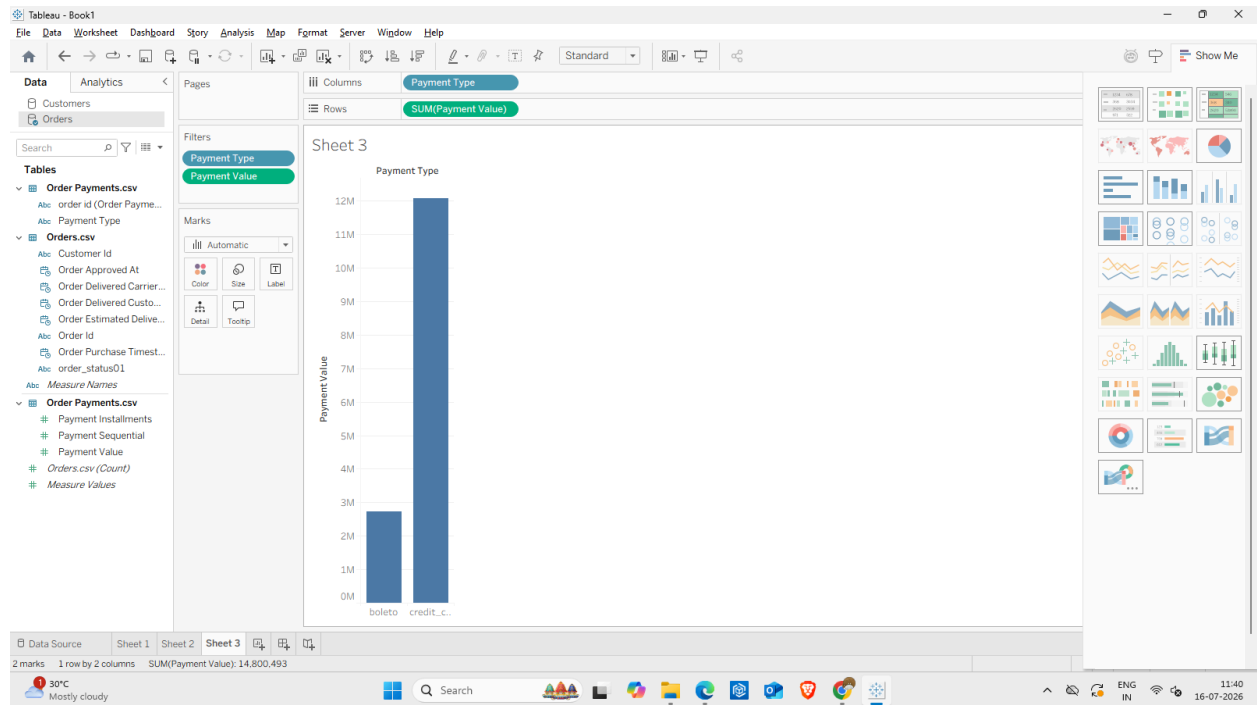
3. Drag **payment_value** to the **Filters** shelf, choose **All Values**, select **At Least**, enter **50**, and click **OK**.



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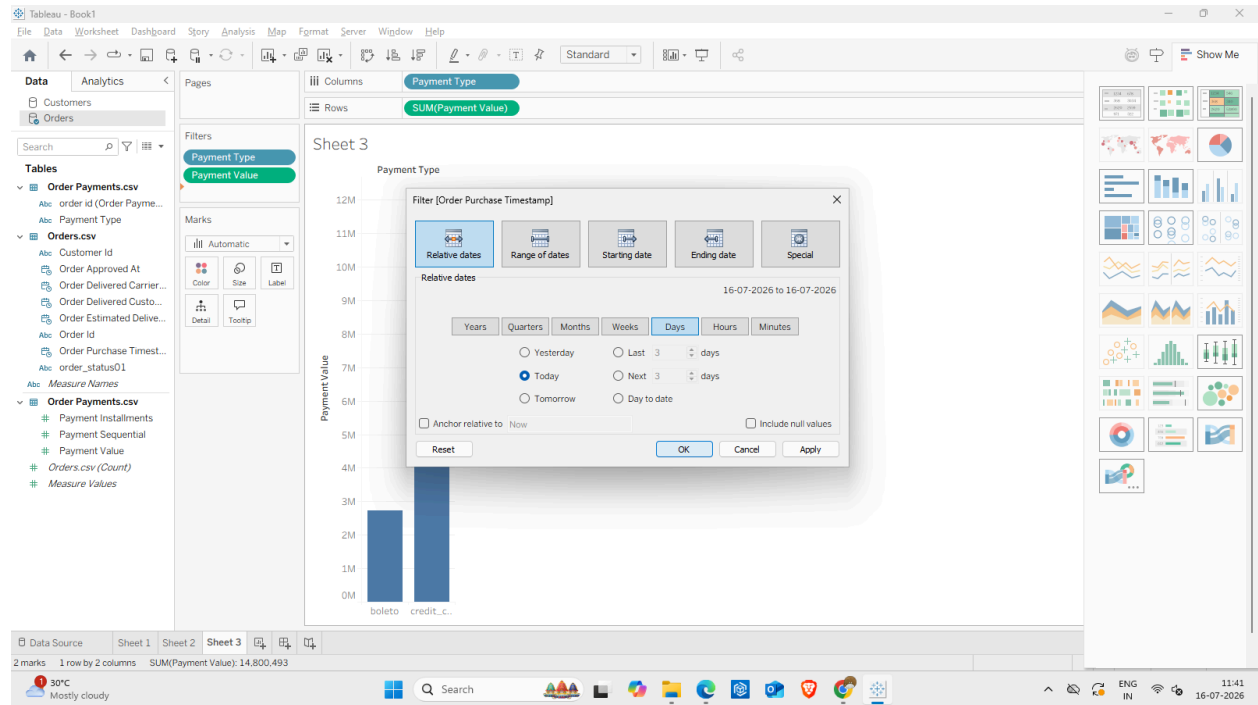
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4. Drag **order_purchase_timestamp** to the **Filters** shelf, choose **Years** or **Relative Date**, and click OK.



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Step G: Optimizing Dashboard Performance

1. Go to the **Data Source** page, click **Edit** under **Extract**, then **Add** the **orders_status01** field. Select only **delivered** and click **OK**.

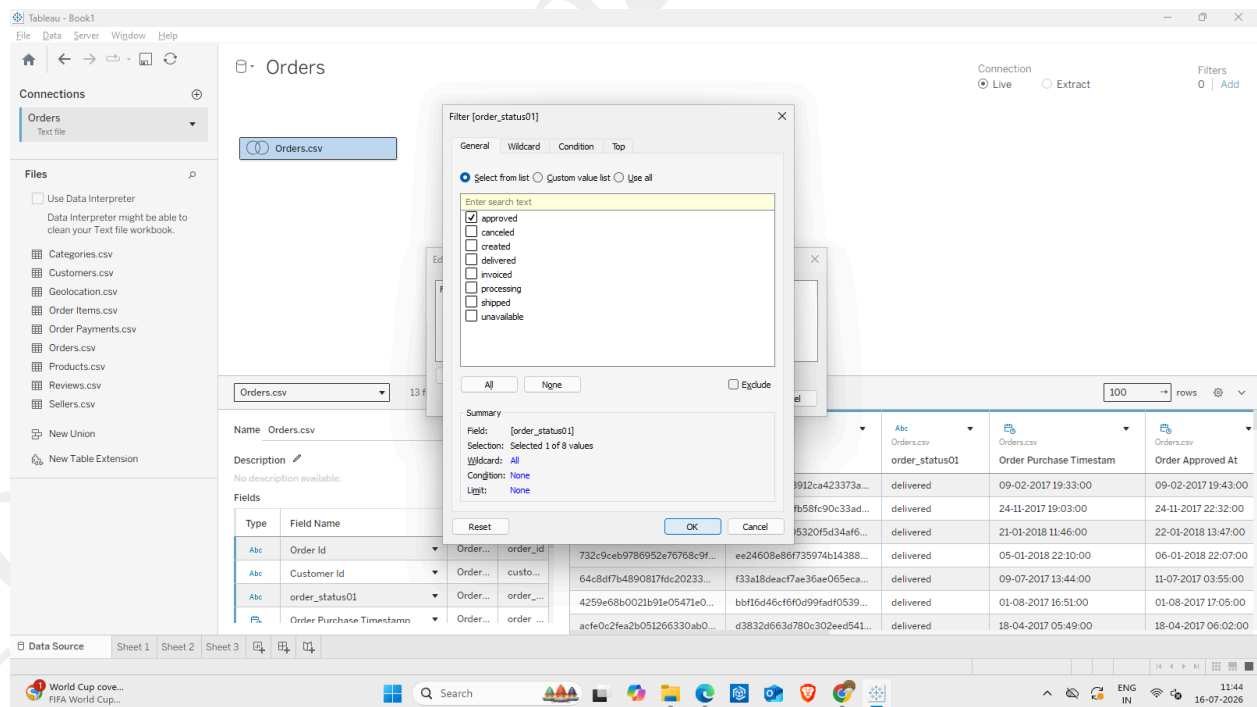
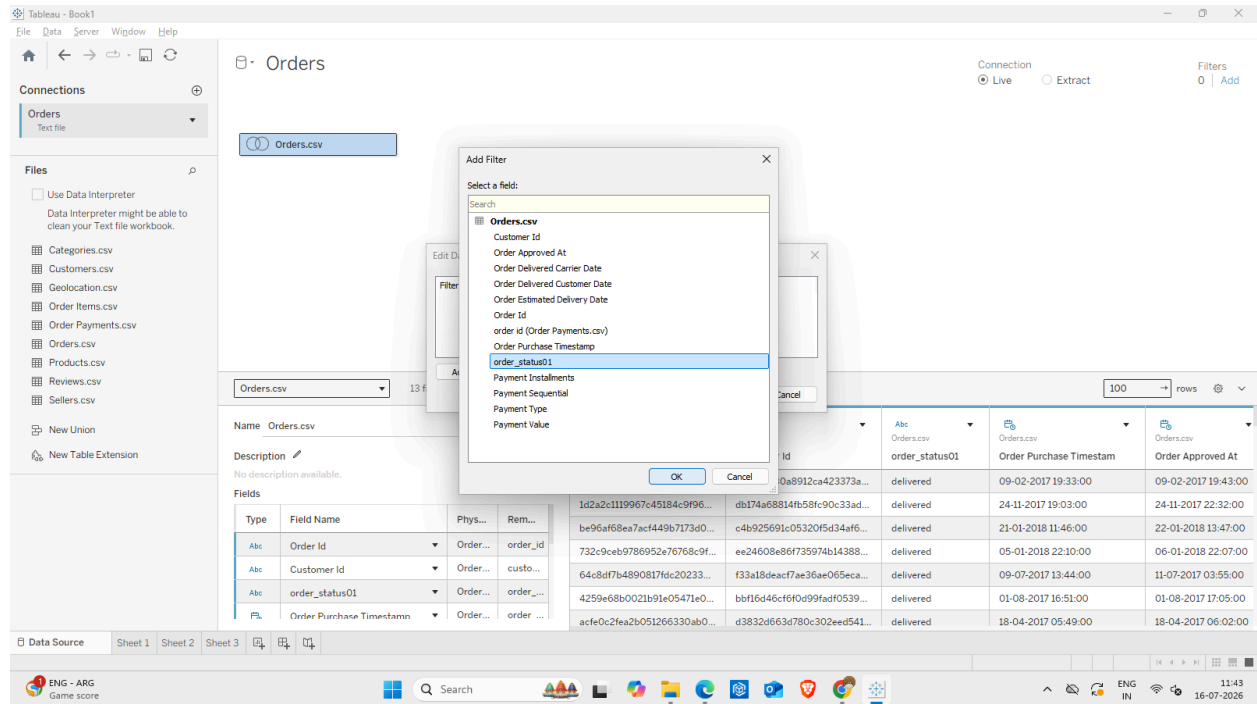
The screenshot shows the Tableau Desktop interface with the 'Orders' connection selected. The 'Filters' button is highlighted in the top right corner. The 'Fields' list on the left includes 'Order Id', 'Customer Id', 'order_status01', 'Order Purchase Timestamp', and 'Order Approved At'. The 'Fields' table is shown below.

Type	Field Name	Phys...	Rem...
Abc	Order Id	Order...	order_id
Abc	Customer Id	Order...	custo...
Abc	order_status01	Order...	order_...
Abc	Order Purchase Timestamp	Order...	order...

The screenshot shows the 'Edit Data Source Filters' dialog box. The 'Filter' tab is selected, and the 'Add...' button is highlighted. The 'Filter' list is empty, and the 'Details' list is also empty. The 'Add...' button is highlighted in the bottom left corner.

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The screenshot shows the Tableau Desktop interface with the 'Orders' data source selected. The 'Edit Data Source Filters' dialog box is open, displaying the 'Filter' tab. The 'Filter' field is set to 'order_status01', the 'Scope' is 'Orders.csv and related tables', and the 'Details' are 'keeps approved'. The 'Add...' button is highlighted. The background shows the 'Orders.csv' data source with 13 fields and 2 rows of data.

Type	Field Name	Phys...	Rem...
Abc	Order Id	Order...	order_id
Abc	Customer Id	Order...	custo...
Abc	order_status01	Order...	order_...
Abc	Order Purchase Timestamp	Order...	order ...

The screenshot shows the Tableau Desktop interface with the 'Orders' data source selected. The 'Orders.csv' data source is displayed with 2 rows of data. The 'Filter' tab is selected in the 'Edit Data Source Filters' dialog box, showing the 'Filter' field set to 'order_status01', the 'Scope' is 'Orders.csv and related tables', and the 'Details' are 'keeps approved'. The 'Add...' button is highlighted. The background shows the 'Orders.csv' data source with 13 fields and 2 rows of data.

Type	Field Name	Phys...	Rem...
Abc	Order Id	Order...	order_id
Abc	Customer Id	Order...	custo...
Abc	order_status01	Order...	order_...
Abc	Order Purchase Timestamp	Order...	order ...

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2. Return to the worksheet, right-click any filter on the **Filters** shelf, and select **Add to Context**. The filter will turn gray.

